

Alessia Pasquini

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[Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

RESEARCH INTERESTS

Sex differences in the brain · Women's health and gender equality · Developmental neuroscience · Cognitive neuroscience · Computational modelling of brain data.

EDUCATION

Ph.D. in Molecular Biology, Biomedicine & Clinical Research (BMBIC) 2024 – present
University of Seville. Doctoral Programme in Molecular Biology, Biomedicine and Clinical Research (BMBIC). Thesis: "Sex differences in the brain in schizophrenia: a normative neuroimaging approach to advancing women's health and equality." Supervisor: Prof. Rafael Romero-García.

M.Sc. in Cognitive Neuroscience & Clinical Neuropsychology 2020 – 2023
University of Padua, Italy. Thesis: "An fNIRS investigation of the perception of non-adjacent linguistic regularities across development". Supervisor: Prof. Judit Gervain. Grade: 110/110 cum laude.

B.Sc. in Cognitive & Psychobiological Sciences 2017 – 2020
University of Padua, Italy. Thesis: "A behavioral and fNIRS investigation of spatial-numerical associations in 7-month-old infants". Supervisors: Prof. Sabrina Brigadoi and Prof. Silvia Benavides-Varela. Grade: 102/110.

RESEARCH EXPERIENCE

Research Assistant — Neuroimaging and Brain Networks Unit 2024 – present
University of Seville, Seville, Spain. Neuroimaging of psychiatric disorders; normative brain-chart modelling of sex and age effects across the schizophrenia spectrum.

- Apply linear and machine-learning models to a global multi-site neuroimaging dataset of ~80,000 subjects to characterize sex- and age-dependent effects in schizophrenia and related disorders.
- Build reproducible analysis pipelines in Python, R and MATLAB using normative (brain-chart) modelling and unsupervised clustering to identify patient subgroups.

Research Assistant — Babylab 2021 – 2022
University of Padua, Padua, Italy. Developmental cognitive neuroscience (fNIRS).

- Designed, ran and analysed a self-conceived neuroimaging study end to end; managed participant recruitment and led an outreach/advertising campaign.
- Collaborated on joint research projects with the University of Paris (France) and the University of British Columbia (Canada).

Research Assistant — Dept. of Developmental & Social Psychology 2019 – 2020
University of Padua, Padua, Italy. Neuroimaging and behavioral data collection.

- Collected neuroimaging (fNIRS) and behavioral (eye-movement) data and designed a standardized data-collection procedure.

- Analysed behavioral data, built appropriate statistical models, and mentored new researchers on the experimental procedure.

PUBLICATIONS

- Alemán-Morillo, C., García-San-Martín, N., Bethlehem, R. A. I., Dorfschmidt, L., Alemany-Navarro, M., Segura, P., **Pasquini, A.**, Muñoz-Caracuel, M., Canal-Rivero, M., Seidlitz, J., Ayesa-Arriola, R., Vázquez-Bourgon, J., Suckling, J., Ruiz-Veguilla, M., Crespo-Facorro, B., & Romero-García, R. (2025). Medication and atypical brain maturation in psychosis associated with long-term cognitive decline and symptom progression. *The British Journal of Psychiatry*, 1–11. <https://doi.org/10.1192/bjp.2025.10482>
- Gemignani, J., de la Cruz-Pavía, I., Martinez, A., Nallet, C., Lucarini, G., Cavicchiolo, F., **Pasquini, A.**, & Gervain, J. (2023). Reproducibility of infant fNIRS studies: a meta-analytic approach. *Neurophotonic*, 10(2), 023518. <https://doi.org/10.1117/1.NPh.10.2.023518>

CONFERENCE PRESENTATIONS AND POSTERS

- **Pasquini, A.** et al. (2026). Mapping sex and age effects using brain charts in schizophrenia spectrum disorder. Poster, *FENS Forum 2026*.
- **Pasquini, A.** et al. (2025). Brain charts, the new pioneer of psychiatry. Poster, *Brain Awareness Week*, Department of Medicine, University of Seville.

TECHNICAL SKILLS

Programming and analysis: Python, R, MATLAB.

Methods: linear and mixed-effects statistical modelling, machine learning, unsupervised clustering, normative (brain-chart) modelling, fNIRS data analysis, reproducible pipelines (Git).

LANGUAGES

Italian (native) · English (C1/C2) · Spanish (C1/C2) · French (B1).

PROFESSIONAL MEMBERSHIPS AND SERVICE

- Member, [ALBA Network](#) (diversity and inclusion in the brain sciences).
- Member, [SABV.eu](#) (sex and gender dimension in biomedical research).